

Ziru Wei

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Keywords: Human-Computer Interaction; Human-Robot Interaction; Context-Aware Interaction

RESEARCH INTERESTS

My research explores next-generation AI interfaces designed to augment humans undistruptively in everyday physical environments. I develop medium-agnostic and context-adaptive systems spanning modalities from extended reality to robotic interfaces, and investigate how humans perceive them.

EDUCATION

- The ATLAS Institute, University of Colorado Boulder** Starting Fall 2026
- PhD in Creative Technology and Design
 - Advisor: Prof. Ryo Suzuki
- Carnegie Mellon University** Graduated May 2026
- Master of Science in Computational Design (research-based)
 - Advisor: Prof. Alexandra Ion
 - Thesis Project: *Toward Interacting with Proactive Intelligence in Everyday Physical Environments*
- Soochow University** Graduated Jun 2024
- Bachelor of Architecture

PUBLICATIONS

- [P1] **While You're Stil Busy: Context-Aware Approach Optimization for Proactive Mobile Robots**
Ziru Wei, Di Wen, Leyuan Chen, Hassan Shahzad, Alexandra Ion *UIST'26* (ACM Symposium on User Interface Software and Technology, 2026), full paper accepted
- [P2] **Objects with Arms: How Task-Relevant Embodiment Shapes Perception in Physical Collaboration**
Violet Yinuo Han, Ziru Wei, Aiden Yiliu Li, Chris Wu, Alexandra Ion *RO-MAN'26* (IEEE International Conference on Robot and Human Interactive Communication, 2026), full paper accepted
- [P3] **Embodied Generative Taskscape: Re-Connect Cultural-Ecological Perception in 5-Animal Play**
Ziru Wei, Jimmy Cheng. *4S'25* (50th Annual Meeting of the Society for Social Studies of Science, 2025) abstract accepted
- [P4] **On-site Holographic Building Construction: A Case Study of Aurora**
Sijie Liu, Ziru Wei, Sining Wang. *CAADRIA'22* (Proceedings of the Association for Computer-Aided Architectural Design Research in Asia, 2022) full paper accepted (~30% acceptance, top-tier conference in computational design)

RESEARCH EXPERIENCES

- Master's Student | Interactive Structures Lab, Carnegie Mellon University** Mar 2025 - Present
- Advisor: Prof. Alexandra Ion
 - Developed software and hardware systems for proactive robotic interfaces that blended into everyday environments, and understood human perceptions of them (See R1, R2, R3).
- Research Intern | WHY Research Lab, School of Architecture, Carnegie Mellon University** Aug 2024 - Jan 2025
- Advisor: Prof. Daragh Byrne
 - Replicated an e-waste scanner by disassembling and re-soldering disk drives, integrating a Raspberry Pi controller; built the WasteStation database in Notion to map components and their potential reuse applications
- Research Intern | Humachine Lab, Architecture Department, Soochow University** May 2021 - Aug 2022
- Advisor: Prof. Sining Wang
 - Designed MR workflows and four on-site collaboration methods for nonlinear façade assembly with limited devices and a small construction team; documented the design-to-construction process and contributed to the literature review part in paper writing (See R5)

AWARDS AND HONORS

- Computational Design Commendation, Carnegie Mellon University** 2025

Computational Design Commendation, Carnegie Mellon University	2024
Merit Scholarship, Carnegie Mellon University	2024
Outstanding Undergraduate Thesis (Top 1%), Soochow University	2024
Excellence Award, Shanghai Youth Architectural Design Competition	2023
Innovation & Academic Excellence Scholarship, Soochow University	2020 - 2022
Overall Excellence Award Winner, Solar Decathlon China	2022
First Prize (Top 2%) in “Zijin Award” of Architectural Design Contest	2022
METTLER TOLEDO Scholarship (Top 2%)	2019

MENTORING

Tanvi Handoo , Undergraduate Student in Computer Science, Carnegie Mellon University	May 2025 - Sept 2025
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ACTIVITIES

Course Guest Reviewer, Carnegie Mellon University	Jan 2025 - May 2025
Student Volunteer, des[AI]gn conference 2024, American Institute of Architecture Students	Oct 2024

SKILLS

Technical

- Hardware: Arduino, Raspberry Pi
- Programming: Python, PyTorch, Rstudio, C#, Pascal, HTML, CSS, JavaScript

Design & Production

- Software: Unity, Rhino, Grasshopper with GHPython, Blender, AutoCAD, Adobe Creative Suite, Figma, Procreate
- Fabrication: 3D printing, Soldering and electronic wiring, Welding (basic), Woodworking

Languages

- English (Fluent), Mandarin (Native), Portuguese (Beginner)